XRPL Validator — Daily Engineering Report · 2026-10-02

Period: 2026-10-02 00:00 to 24:00 US Eastern (2026-10-02 04:00 to 2026-10-03 04:00 UTC); ledgers #107,373,862 to #107,396,150. Every count from the ledger and the validator's logs stops at #107,396,150. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,289 of the 22,289 ledgers the network closed (100.000%) (agreement not checked: the ledgers' hashes are no longer held).
- **Native engine:** 1 engine receipt recorded (1 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 0 deployments, 1 commit merged into main, 1 engine finding fixed (F411).

Service levels

Measure	Value
Ledgers closed by the network	22,289
Validations issued	22,289
Coverage (validations / ledgers closed)	100.000%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	0
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	0
Basis: the hash the first peer announced as accepted (before cycle 183)	22,289
External agreement, data.xrpl.org (UTC day 2026-10-02)	0.99196 (15 missed of 1,865)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,373,862–#107,396,150), by step:

Step	Ledgers with a receipt
Step 1 · native engine	1
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,381,776	Step 1 · native engine	26 ledger change(s) the network made were absent from ours: 27144336 (deleted), FF3EA0AC (deleted), 362AF11C (deleted); 8 object(s) differed in content	fixed: F411, commit 4729eb4, deployed in cycle 164 (2026-10-03 12:50 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-02-01, INC-01 is INC-2026-10-02-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

No incidents.

Change log

Deployments

No deployments.

Merged into main

Merged	Repo	Commit	Subject
12:51 UTC	xrpl-validator	`4729eb4`	fix(xrpl-ledger): F411 — the flow's dirFirst reads a relinked page link

Findings fixed

- **fix(xrpl-ledger): F411 — the flow's dirFirst reads a relinked page link** (`4729eb4` , xrpl-validator) Mainnet #107381776 DC4F4F76FD26: rGqo7tmV's tfSell|tfIoC sells 1,004.58 RLUSD for at least 647.04 XRP. One quality level of the RLUSD/XRP book spans three pages (32, 32 and 16 offers). The cross empties the root page, then page 1; deleting page 1 relinks the root's IndexNext to page 2. directory.rs `set_link` writes that link as a JSON number, but the flow's `dir_first` read IndexNext only as a hex string, so the …

Independence program

No independence record for UTC day 2026-10-02; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-02 12:51	10-02 12:51	xrpl-validator	`4729eb4`	fix(xrpl-ledger): F411 — the flow's dirFirst reads a relinked page link

Times in UTC. A commit appears on the day it was committed to main or merged into it.